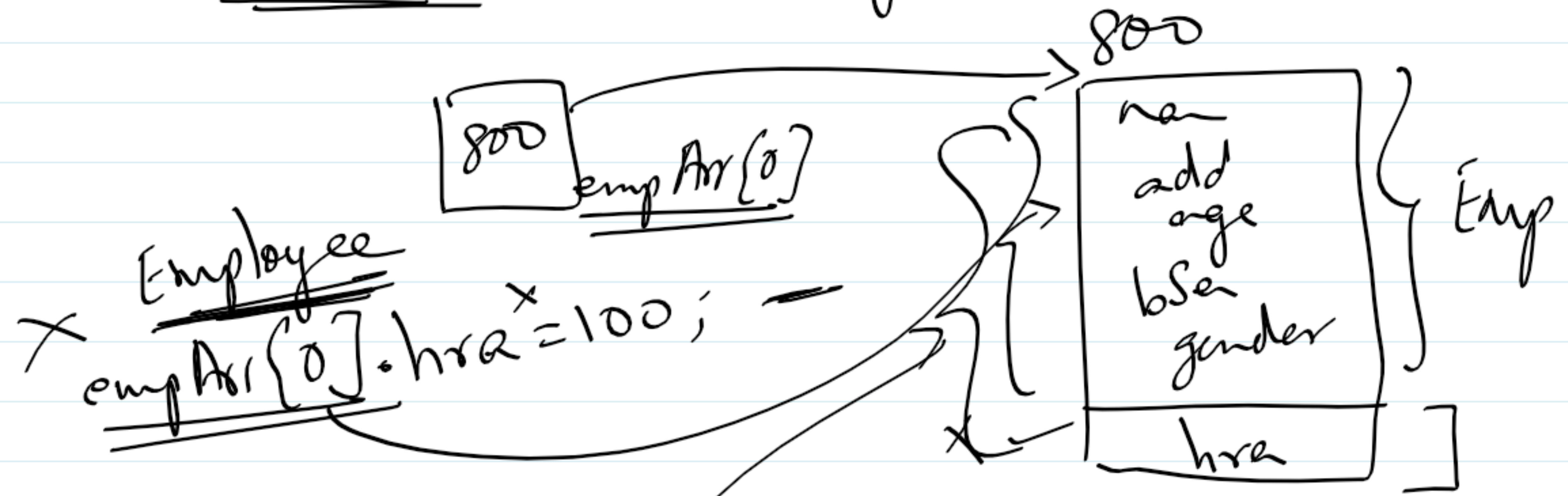
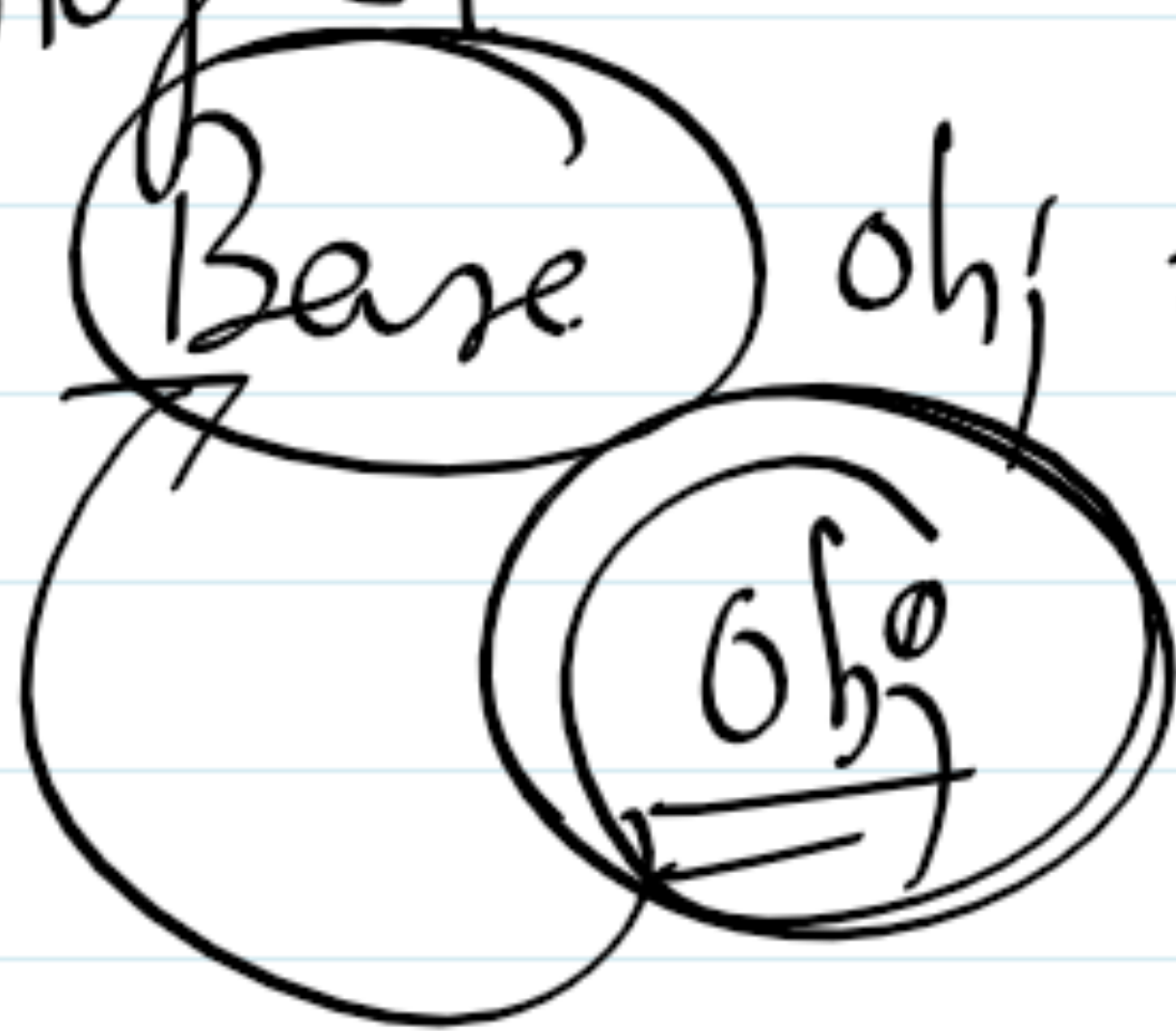
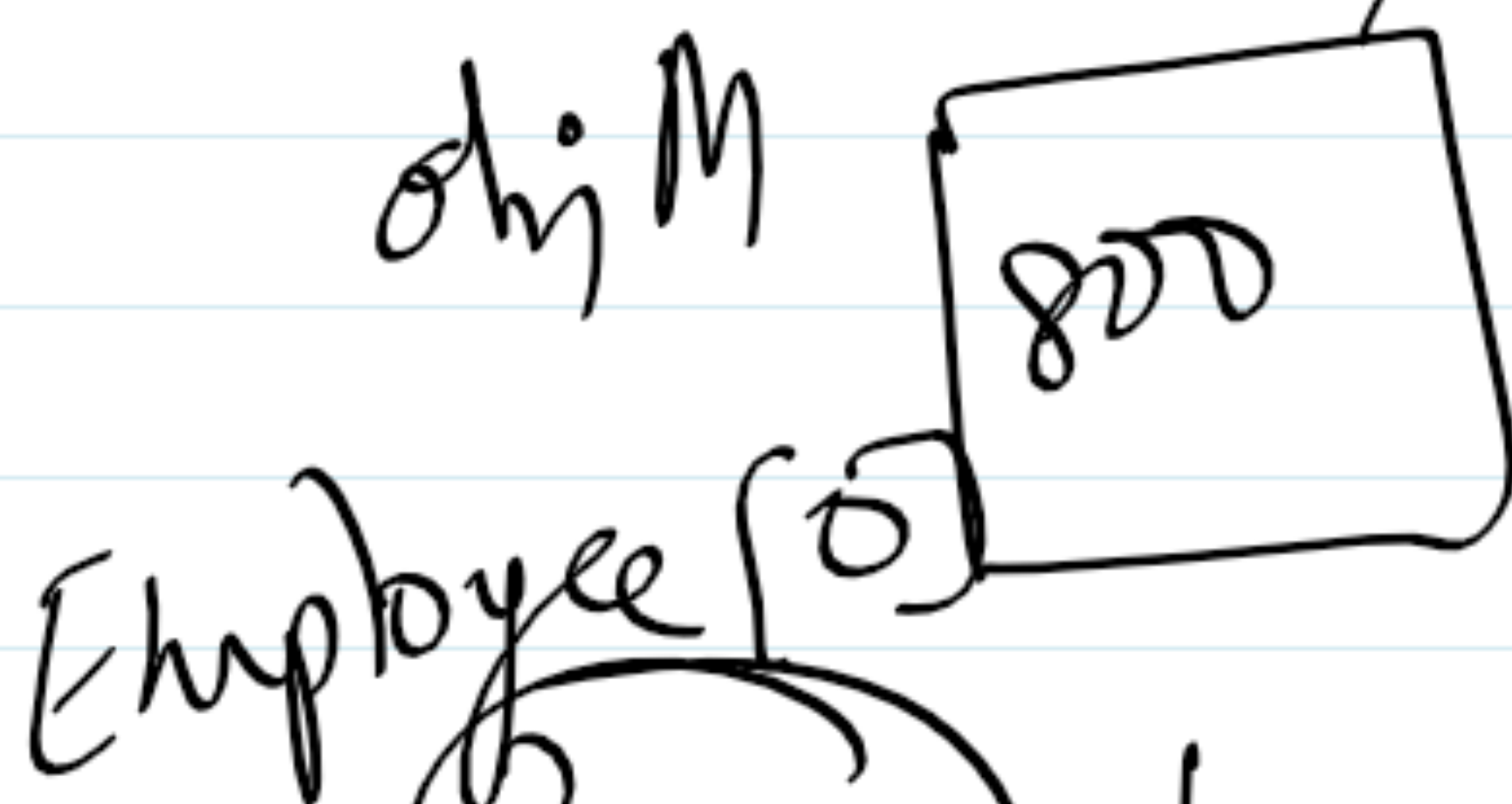


empArr[0] = new Manager(---);



Manager objM = (Manager) empArr[0];

objM.hra = 5



obj = new Demind ();

<u>Microsoft</u>	→	Oracle
{ ① <u>Calculator</u> → <u>Sci</u> { ② <u>Date</u>		✓ ① <u>Calculator</u> - <u>Basic</u> ✓ ② <u>Date</u> 10-15 ① :

X Date obj Date = new Date();

Ambiguity



Packages

Classes

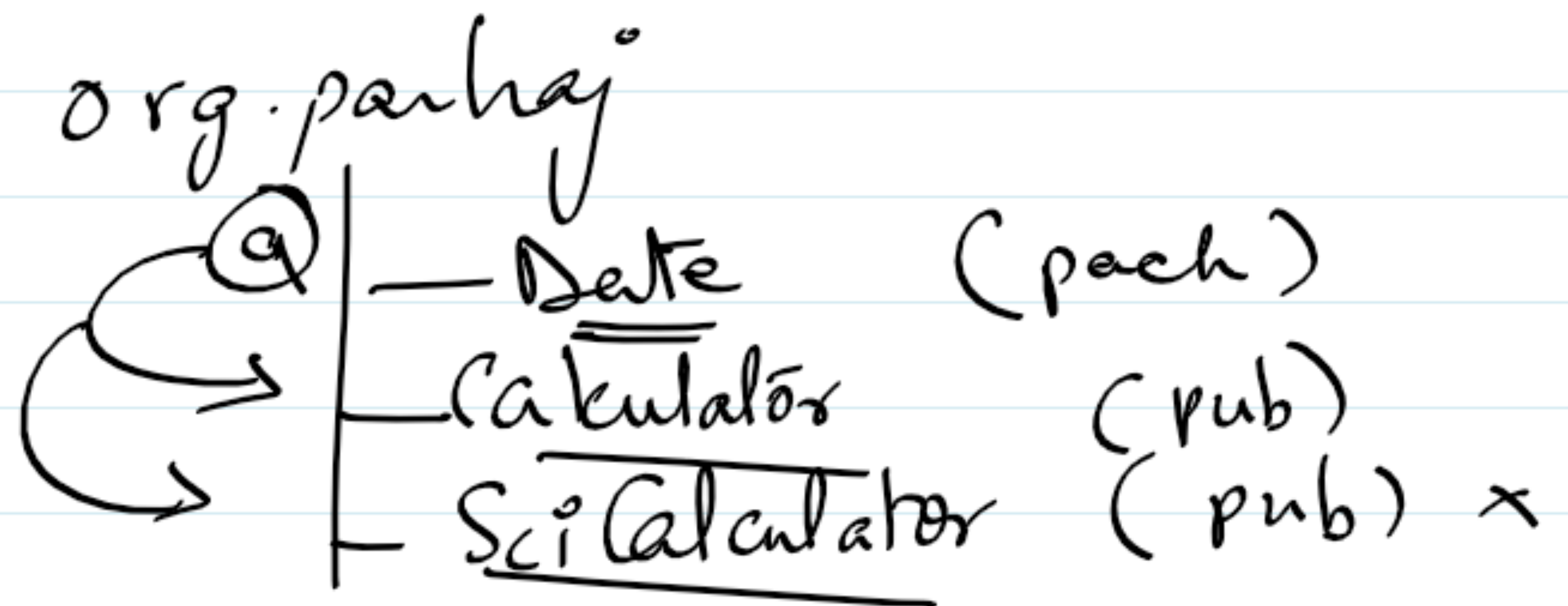
encapsulate

package org.panhaj;

o.p.v.date
 in .cdac.math
 in .cdac.chem
 in .ch --

public class Date

X class Date (package scope)



package in.cdac;

pub class Progr

main()

3 Calculator / SciCal

✓

```

package ✓ in .cdac;
import org.pandaj.*; a) Date;
class Progr b) *;
{

```

```

    main ( )
    {
        ✓ Date obj = new Date();
    }

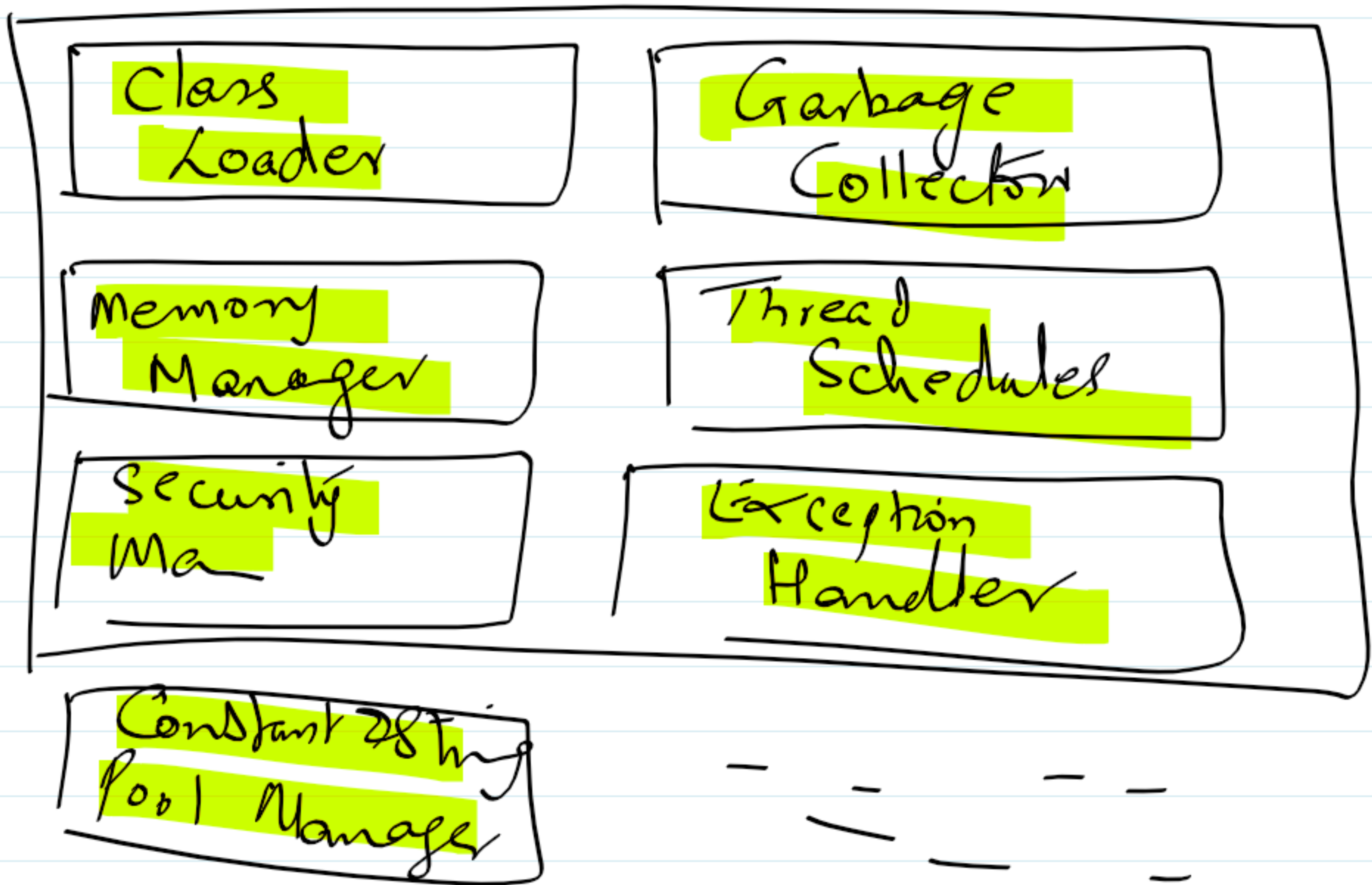
```

```

package in .cdac;
import org.pandaj.*;
class SpecialDate extends Date o.p
{
}

```

JVM



① Class Loader (Dynamically)

• class

① 1st time an object of this class created
OR//

② 1st time a static variable / function is
accessed from a class.
OR//

③ Programmer explicitly loads the class
in memory.
`Class.forName("in.cdac.Base");`

Class Loader loads class.

- ① allocates & initializes all static data members of the class.
 - ② invokes the static initialization block of this class
-

C++

```
class Data
{
    int n1;
    float n2;
};
```

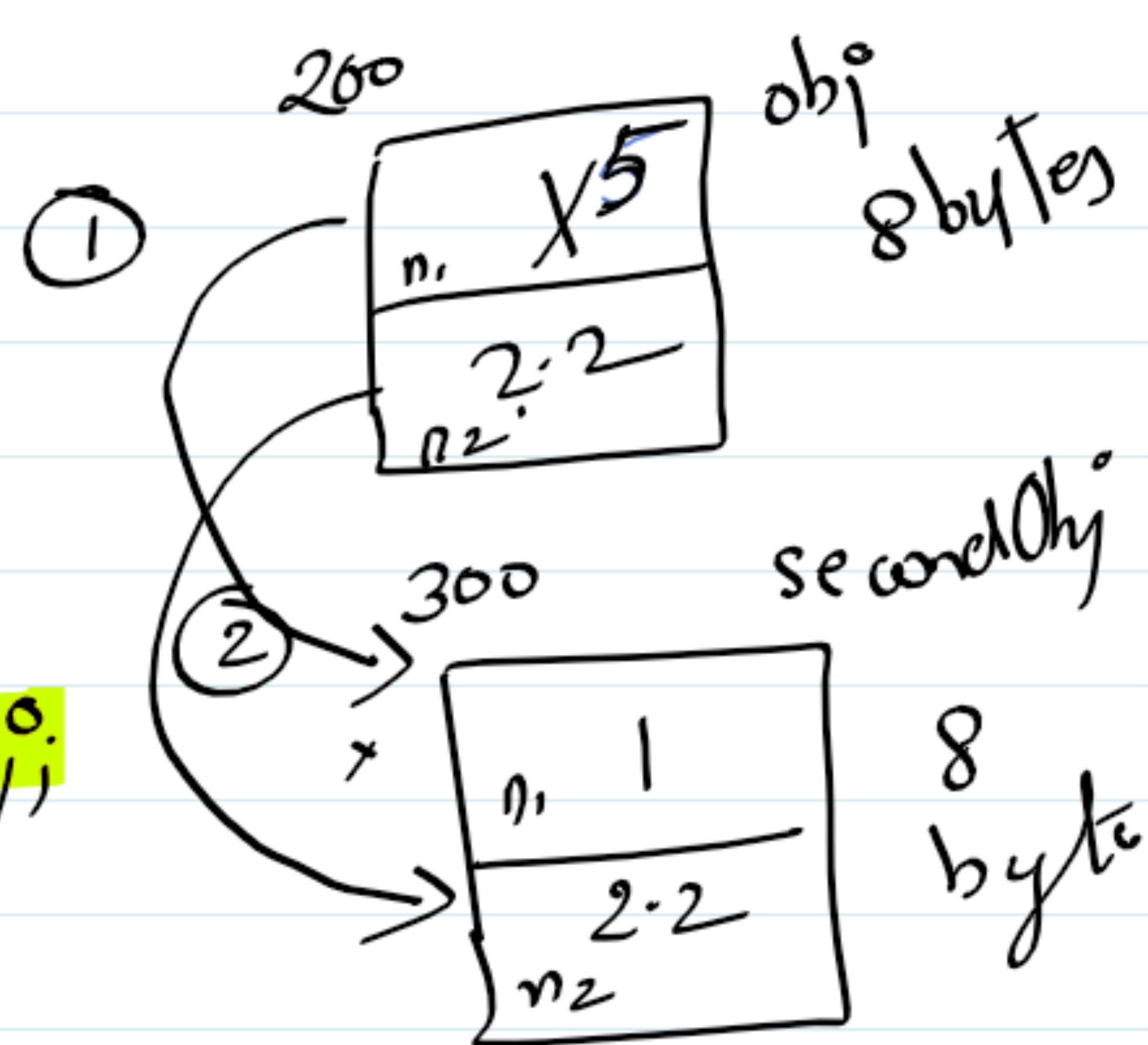
① Data Obj;
obj.n1 = 1;
obj.n2 = 2.2;

② Data secondObj = obj;

Copy Ctor

✓ obj.n1 = 5;

→ secondObj.n1 = 1

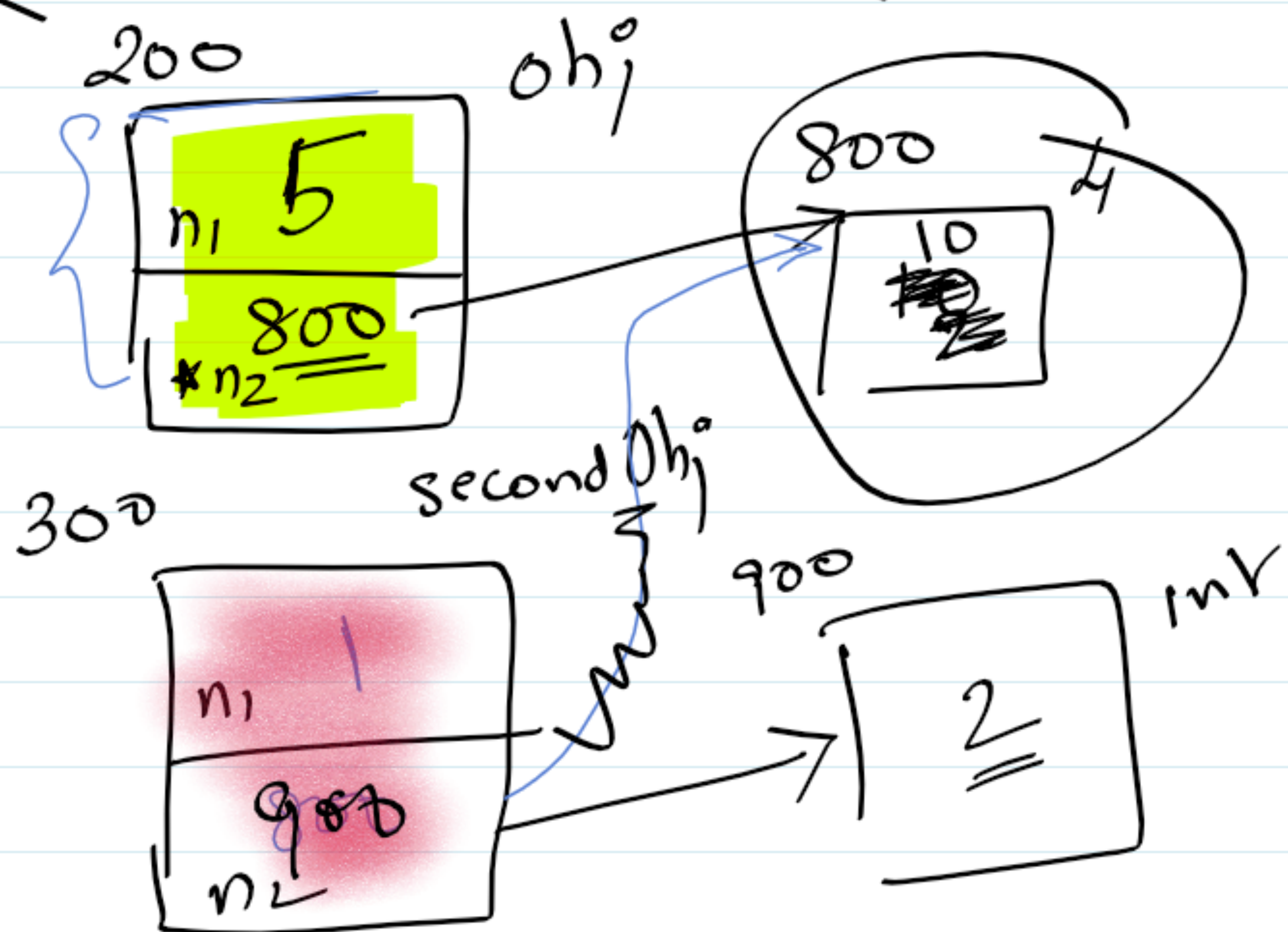


```
class Data
{
    int n1;
    int *n2;
};
```

```
Data obj;
obj.n1 = 5;
obj.n2 = new int;
*obj.n2 = 2;
```

Data secondObj = obj;

obj.n1 = 5;
*obj.n2 = 10;



secondObj.n1
*secondObj.n2 → ?

internal Copy Ctor

secondObj.n1 = obj.n1
n2 = obj.n2

Deep Copy

secondObj.n1 = obj.n1
secondObj.n2 = new int;
*so.n2 = *o.n2;

Java

char → 2 bytes → Unicode

"Hello"
C++ = 5 bytes

Java 10 bytes

```
{ String str = "Hello" 10  
 3 ← 1 X : 200  
  String str1 = "Hello"; 10  
  Sys.out.println("Hello"); 10
```

String Pool

Content	Address
"Hello"	200
"Bye"	300
...	...

String str = "Hello";

200
1000 str

String str1 = "Hello";

200
1100 str1

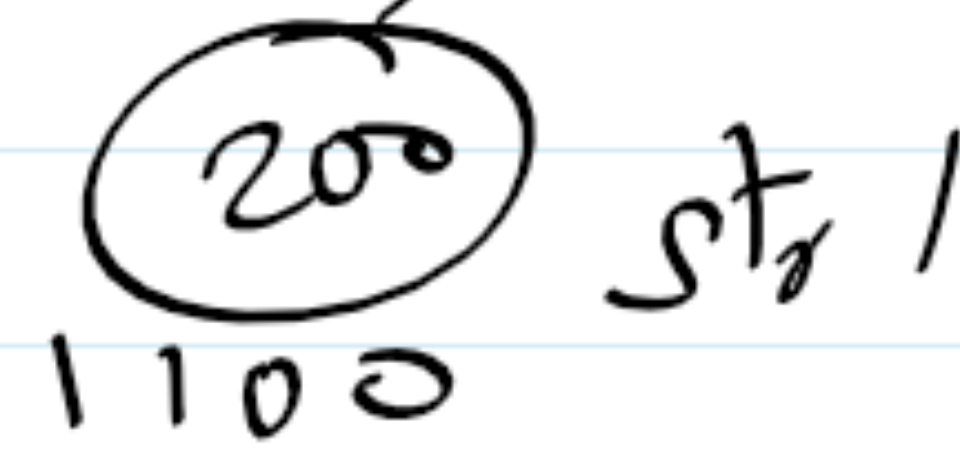
str = "Bye"; str1

String in Java are immutable not changeable

String str = "Hello";



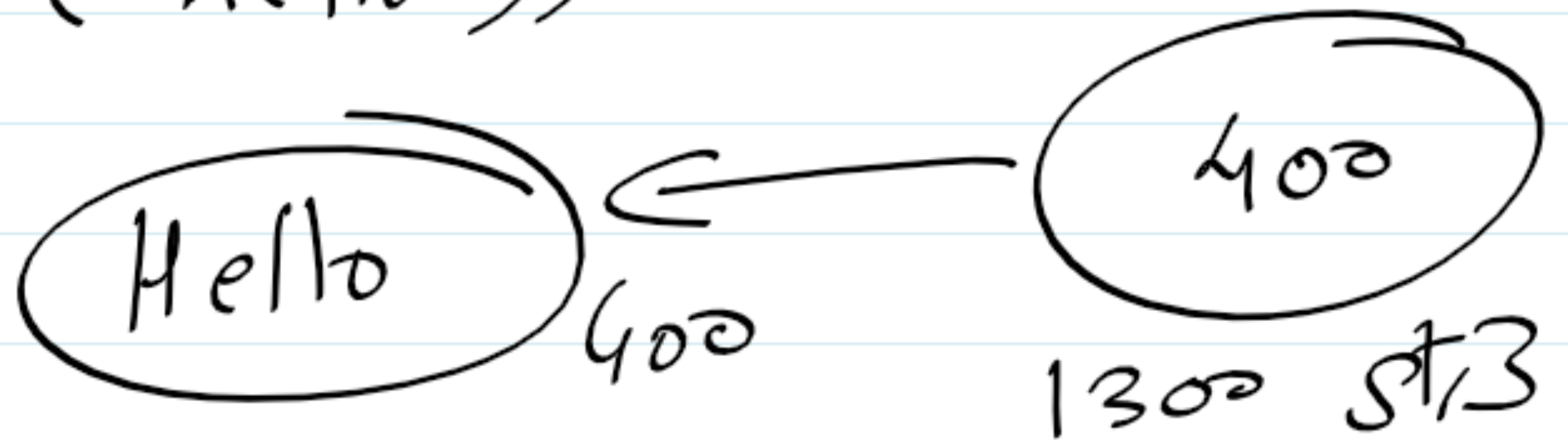
String str1 = "Hello";



String str2 = "hello";



String str3 = new String("Hello");



String str = "Hello" + "Bye" + "Hello" + "Bye";

Diagram showing the concatenation of strings "Hello" and "Bye" into "HelloBye".

Mutable

StringBuffer

1 - Hello
2 - Bye
3 - HB
4 - HBH
5 - HBHB

"Hello" 5 - HBHB

StringBuffer buffer = new SB("Hello");

buffer.append("Bye");
buffer.append("Hello");
buffer.append("Bye");

HBHB

String str = buffer.toString();